Experiment 6 - Day 3: Conversion of Camphene to Camphor

Hannah Owens

11/28/2023

CHEM 2081 - 003 - 402

Introduction

In this experiment, camphene, acetone, dichloromethane, acetic acid, aqueous solution of sodium bicarbonate, sodium hypochlorite, borneol, and camphor were used to convert borneol to camphor. The purpose of the entire experiment is to produce camphor from camphene, but in day 3, camphor is produced from borneol.

Methods and Materials

In a long-neck 10 mL flask, 0.3 g of borneol product was added along with 0.3 mL of glacial acetic acid. The mixture was dissolved in 1 mL of acetone and a boiling chip was added. A reflux was assembled with the distillation column and the apparatus was placed in a water bath with the temperature sitting between 40 and 50. Once the reagents are dissolved, 6 mL of aqueous sodium hypochloride was added. This was done slowly and the mixture was swirled. The reaction was allowed to continue for 15 minutes. After completion, the mixture was cooled to room temperature and the camphor was extracted with two portions of methylene chloride. The combined methylene chloride layers was neutralized with ~2 mL of saturated sodium bicarbonate. Bicarbonate was added until the presence of acidic residuals were gone. The organic solution was washed with 2 mL of 5% sodium bisulfite and 2 mL of water. The organic solution was dried over anhydrous sodium sulfate. Lastly, the DCM was evaporated over a water bath. The collected crystals were weighed and the percent yield was determined.

Results

Watch glass without sample (g)	35.866 g
Watch glass with sample and drying agent (g)	36.232 g
Drying agent added (g)	0.3 g
Sample (g)	0.066 g

Conclusion

In this experiment, camphene, acetone, dichloromethane, acetic acid, aqueous solution of sodium bicarbonate, sodium hypochlorite, borneol, and camphor were used to convert borneol to camphor. In day 3, borneol is used to produce camphor. The final dried sample was 0.066 g. In step 7 and 8, the anhydrous sodium sulfate was added and not removed before the DCM was added. This was a major source of error. After evaporating, the sample and drying agent were undistinguishable. Another source of error could be that the aqueous sodium hypochloride solution was added too quickly and not mixed thoroughly enough.

References

H McNeil. Extraction. *Chem 2081 Organic Chemistry II Lab Manual*, Macmillan Learning Curriculum Solution: Plymouth, 2023, (p. 69-72).

Exp. No. 6-D3	Experiment/Subject Conversion of Camphene to Camphor	Date 11.14.23	
Name Hannah Owens	Lab Partner	Locker/ Desk No.	Course & Section No. 2081 003

Procedure	Observations
1. In a long-neck 10 mL round-bottom flask, add racemic 0.3 g of borneol product with 0.3 mL of glacial acetic acid. Dissolve the racemic mixture in 1 mL of acetone. Add a boiling chip.	1. Added 0.3 g borneol to 0.3 mL glacial acetic acid. Dissolved in 1 mL acetone.
2. Assemble a reflux with the distillation column and place in a water bath with temperature 40-50 °C. Don't let the temperature deviate. If the borneol doesn't dissolve add 0.5 mL of acetone to the mixture until all borneol goes in the solution.	2. Assembled reflux
3. Once all reagents dissolve add 6 mL of aqueous sodium hypochloride solution (bleach). The addition should be done through the top opening of the condenser and should be paced dropwise over 35 minutes. After each addition, the mixture should be swirled gently. The reaction must continue for 15 minutes after the last bleach addition.	3. Added 6 mL bleach slowly and reaction proceeded for 15 minutes
4. Let the mixture cool to room temperature and extract the camphor with 2 portions (2mL each) of methylene chloride. Use a centrifuge tube.	4. Cooled to room temperature
5. Neutralize the layers with ~2mL of saturated sodium bicarbonate. Keep adding if necessary. The presence of acidic residuals will be determined by the production of CO_2 bubbles after every addition of the bicarbonate solution.	5. Layers were neutralized until acidic residuals were gone.

Signature Hannah Owens	Date 11.14.23	Witness/TA	Date

Note: Place fold-over back cover under copy sheet before writing

Exp. No. 6-D3	Experiment/Subject Conversion of Camphene to Camphor	Date 11.14.23	
Name Hannah Owens	Lab Partner	Locker/ Desk No.	Course & Section No. 2081 003

Procedure	Observations
6. Wash the organic solution sequentially with 2 mL of 5% sodium bisulfite and with 2 mL of water.	6. Organic solution was washed
7. Dry the organic solution over anhydrous sodium sulfate. The solution is "dry" when freely rolling crystals of the drying agent are observed.	7. The solution was dried over anhydrous sodium sulfate
8. Evaporate the DCM over a wath bath with a temperature below 50°C. Weigh the collected crystals and determine percent yield.	8. DCM evaporated over water bath

Signature Hannah Owens	Date 11.14.23	Witness/TA	Date

Note: Place fold-over back cover under copy sheet before writing

Lab Performance

Exp 6 D3

Hannah Owens | 003 | 402

Name: | *Section:* | *Room:* | *Hood:*

Wearing personal protection equipment—this evaluation records your consistency with wearing the appropriate clothes for lab (long pants, closed toe shoes, and tops with sleeves) as well as the required laboratory safety equipment of apron, goggles, and gloves.

○ ○ ○ ○ ~~○~~ (5pts)

Safe working habits—as the GTAs do the lab rotations, they will record your work ethic and whether you follow our safety rules and regulations concerning tied hair, tied apron, safety goggles on at all times, and no food and drinks.

○ ○ ○ ○ ~~○~~ (5pts)

Following our laboratory safety technique work procedures—evaluation of operation of our laboratory equipment—water sinks in hood (1pt), vacuum and air lines (1pt), sash position (1pt), and heating elements (1pt). It is important that after every experiment you shut down the hood and make it ready for the section after yours (5pts).

○(1pt) ○(1pt) ○(1pt) ○ (1pt) ~~○~~ (5pts)

Glassware operation—safe handling of our glassware equipment. This means cleaning (1pt), maintaining the drawer inventory (5pts), and proper experimental setup (5pts).

○(1pt) ○ (5pts) ○ (5pts)

Working independently or equal partner participation during the entire lab.

○ ○ ○ ○ ○ (5pts)

Good work behavior—you work in a dynamic environment; therefore, creating distractions through noise and disobedience will not be tolerated. No cell phone use, listening to loud music, or working/distracting at a non-assigned workspace.

○ ○ ○ ○ ○ (5pts)

Maintain clean and organized personal space.

○ ○ ○ ○ ~~○~~ (5pts)

GTA notes:

GTA signature: _____ *Final Performance Grade:* [ ]